

Esercizi fisica capitolo 18:

$$24) m_{\text{mol}} = 50 \text{ g/mol} = 0,05 \text{ kg/mol}$$

$$Q = 314 \text{ J}$$

$$m = 30 \text{ g} = 0,03 \text{ kg}$$

$$T_i = 25^\circ \text{C}$$

$$T_f = 45^\circ \text{C}$$

$$a) Q = c \cdot m \cdot \Delta T \Rightarrow 314 \text{ J} = c \cdot (0,03 \text{ kg}) \cdot (20^\circ \text{C})$$

$$\Rightarrow c = \frac{314 \text{ J}}{(0,03 \text{ kg})(20^\circ \text{C})} = 523,3 \text{ J/kg} \cdot ^\circ \text{C}$$

$$b) n = \frac{m}{m_{\text{mol}}} = \frac{\text{kg}}{\frac{\text{kg}}{\text{mol}}} = \frac{\text{kg}}{\text{kg}} \cdot \text{mol} = \frac{0,03 \text{ kg}}{0,05 \text{ kg}} \cdot \text{mol} = 0,6 \text{ mol}$$

$$Q = c_n n \Delta T \Rightarrow 314 \text{ J} = c_n (0,6 \text{ mol}) \cdot (20^\circ \text{C})$$

$$\Rightarrow c_n = \frac{314 \text{ J}}{0,6 \text{ mol} \cdot 20^\circ \text{C}} = 26,2 \text{ J/mol} \cdot ^\circ \text{C}$$

$$25) T_i = 0^\circ \text{C}$$

$$T_f = 37^\circ \text{C}$$

$$m = 454 \text{ g} = 0,454 \text{ kg}$$

$$Q = 3500 \text{ kcal} = 3,5 \cdot 10^6 \text{ cal}$$

$$m_{\text{acqua}} = \frac{Q}{c \Delta T} = \frac{3,5 \cdot 10^6 \text{ cal}}{1 \cdot 10^3 \frac{\text{cal}}{\text{kg}} \cdot 37^\circ \text{C}} = \frac{3,5 \cdot 10^6}{37} = 94,6 \text{ g}$$

$$m = 94,6 \cdot 10^4 \cdot \left(\frac{1 \text{ e}}{1000 \text{ g}} \right) = 94,6 \text{ e}$$

27) $m = 130 \text{ g} = 0,13 \text{ kg}$

$$T_i = 15^\circ \text{C}$$

$$T_f = 1235^\circ \text{K} = 961,85^\circ \text{C}$$

$$Q = m c \Delta T = 0,13 \text{ kg} \cdot 236 \frac{\text{J}}{\text{kg} \cdot \text{K}} \cdot 947^\circ \text{K} = 29054 \text{ J}$$

$$Q_L = L \cdot m = 105 \frac{\text{kJ}}{\text{kg}} \cdot 0,13 \text{ kg} = 13,65 \text{ kJ} = 13650 \text{ J}$$

$$Q_{\text{tot}} = Q + Q_L = (29054 + 13650) \text{ J} = 42704 \text{ J}$$

31) $T_i = 100^\circ \text{C}$

$$m_g = 130 \text{ g} = 0,13 \text{ kg}$$

$$T_f = 50^\circ \text{C}$$

(m_{repose})

$$Q_{\text{tot}} = Q_{\text{specifico}} + Q_{\text{lat}}$$

$$Q_{\text{tot}} = m_g c_e \cdot \Delta T = m_g \cdot c_e \cdot (50^\circ - 0,0)^\circ \text{C} + L_f \cdot m_g$$

$$Q_{\text{tot}} = m_r c_e \cdot \Delta T = m_r c_e \cdot (100^\circ \text{C} - 50^\circ \text{C}) + L_v \cdot m_r$$

$$m_g c_e 50^\circ \text{C} + L_f m_g = m_r c_e \cdot 50^\circ \text{C} + L_v \cdot m_r$$

$$\underline{m_g c_e 50^\circ \text{C} + L_f m_g = m_r}$$

$$m_r = \frac{0,15 \text{ kg} \cdot 4187 \text{ J/kg} \cdot ^\circ\text{C} + 333 \cdot 10^3 \text{ J/kg} \cdot 0,15 \text{ kg}}{4187 \text{ J/kg} \cdot ^\circ\text{C} \cdot 50^\circ\text{C} + 2256 \cdot 10^3 \frac{\text{J}}{\text{kg}}}$$

$$= \frac{31402.5 \text{ J}}{2465350 \frac{\text{J}}{\text{kg}}} = 0.013 \text{ kg} \approx 13 \text{ g}$$

35) $V = 130 \text{ cm}^3 \Rightarrow m_c = 0.13 \text{ kg}$

$\bar{T}_i = 80^\circ\text{C}$

$m_g = 12 \text{ g}$ (punto di fusione $= 0^\circ\text{C}$)

$Q_{\text{tot},g} = m \cdot L_f = 0,012 \text{ kg} \cdot 333 \cdot 10^3 \text{ J/kg} = 3996 \text{ J}$

$Q_{\text{spe},g} = m c_g \cdot \Delta \bar{T} = 0,012 \text{ kg} \cdot 4187 \frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}} \cdot \bar{T}_f =$

$= 50,24 \cdot \bar{T}_f$

Condizione di equilibrio $Q_{\text{caffe}} = Q_{\text{tot}}$ e $Q_{\text{caffe}} < 0$

$$m_c \cdot c_e (\bar{T}_i - \bar{T}_f) = 3996 \text{ J} + 50,24 \bar{T}_f$$

$$- m_c \cdot c_e \cdot \bar{T}_f - 50,24 \bar{T}_f = 3996 - m_c c_e \bar{T}_i$$

$$544,3 \bar{T}_f + 50,24 \bar{T}_f = -3996 + 43544.8$$

$$\bar{T}_f = \frac{39548.8}{594.54} = 66.5^\circ\text{C}$$

36) $m_r = 150 \text{ g} = 0.15 \text{ kg}$

$$m_e = 220g = 0,22g$$

$$T_i = 20^\circ C$$

$$T_f = 100^\circ C$$

$$m_{r,2} = 300g = 0,3 kg$$

$$m_{e,2} = 5g = 0,005 kg \text{ (Boiling, } T = 100^\circ C)$$

$$Q_{\text{tot},e} = m_{e,2} \cdot L_v = 0,005 kg \cdot 2256 \cdot 10^3 J/kg = 11280 J$$

$$Q_{\text{spec},e} = (m_e) c_e \cdot \Delta T = 0,22 \cdot 4187 \cdot 80 = 73691,2$$

$$Q_{\text{tot},e} = 11280 + 73691,2 = 85511,2 J$$

$$Q_{\text{cistola}} = m_{c,r} \cdot \Delta T = 0,15 \cdot 306 \cdot 80 = 4632 J$$

$$- Q_{\text{cilindro}} = Q_{\text{cistola}} + Q_{\text{tot},e}$$

$$Q_{\text{cilindro}} = - Q_{\text{cistola}} - Q_{\text{tot},e}$$

$$m_{r,2} \cdot c_r \cdot \Delta T = - 90143,2 J$$

$$\Delta T = \frac{- 90143,2}{115,8}$$

$$T_i = \frac{- 90143,2}{115,8} + 100^\circ C \approx 874,4^\circ C$$

$$(37) m_e = 500g = 0,5 kg$$

$$m_g = 0,5 kg \text{ (punto di fusione)}$$

$$T_i = 90^\circ C$$

$$Q_{\text{let},g} = 0,5 \text{ kg} \cdot 333 \cdot 10^3 \frac{\text{J}}{\text{kg}} = 166500 \text{ J}$$

$$Q_{\text{spe},g} = m_g \cdot c_e \cdot \bar{T}_f =$$

$$Q_{\text{te}} = m_e \cdot c_e \cdot \bar{T}_f - m_e \cdot c_e \cdot \bar{T}_i$$

$$m_e c_e \bar{T}_f - m_e c_e \bar{T}_i + m_g L_f + m_g c_e \bar{T}_f = 0$$

$$\bar{T}_f (m_e c_e + m_g c_e) = m_e c_e \bar{T}_i - m_g L_f$$

$$\bar{T}_f = \frac{m_e c_e \bar{T}_i - m_g L_f}{m_e c_e + m_g c_e} = \frac{188415 - 166500}{4187} = 5.2^\circ \text{C}$$

$\bar{T}_f > 0$, quindi il ghiaccio si scioglie completamente

$$\text{Se } \bar{T}_i = 70^\circ \text{C} \Rightarrow \bar{T}_f = \frac{m_e c_e \bar{T}_i - m_g L_f}{4187} = \frac{-19955}{4187} = -4.8^\circ \text{C}$$

Se $\bar{T}_f < 0$ allora il ghiaccio non si è sciolto completamente, e le temperature di equilibrio è

$\bar{T}_f = 0^\circ \text{C}$, allora

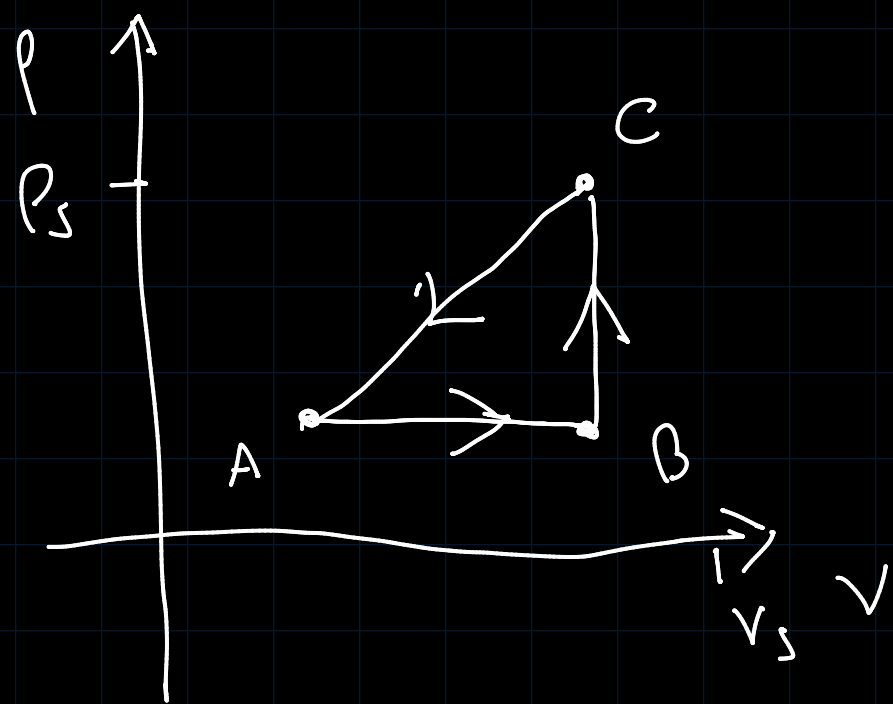
$$0 = c_e m_e \bar{T}_f - c_e m_e \bar{T}_i + L_f m'_g$$

$$m'_g = \frac{c_e m_e \bar{T}_i - c_e m_e \bar{T}_f}{L_f} = \frac{c_e m_e \bar{T}_i}{L_f} = 0.44 \text{ kg}$$

$$m_f = m_g - m'_g = 0.06 \text{ kg}$$

44) $p_s = 40 p_0$ $\Delta E_{int} = Q - L$
 $V_s = 4 m^3$

	Q	L	ΔE_{int}
A $\rightarrow$ B	+	+	+
B $\rightarrow$ C	+	0	+
C $\rightarrow$ A	-	-	-



Nota: Da A $\rightarrow$ B p è costante, la trasformazione è isobara, $L = p \Delta V$

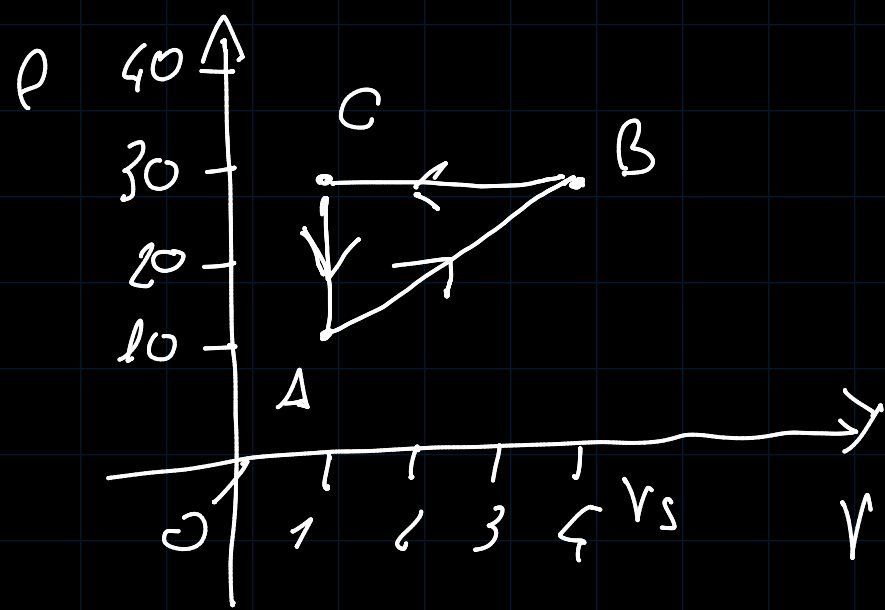
Da B $\rightarrow$ C V è costante, la trasformazione è isocora, $L = 0$

Per la trasformazione C $\rightarrow$ A, il sistema si contrae, quindi $L < 0$
 $Q < 0$
 $\Delta E_{int} < 0$

$$L_{totale} = \frac{1}{2} \text{base} \cdot \text{altezza} = \frac{1}{2} (3-1)(40-20) = 20 J$$

45) $V_s = 4 m^3$

Il ciclo è ABCA.



Durante la trasformazione A $\rightarrow$ B il sistema si espande quindi $L > 0$, $Q > 0$ e $\Delta E_{int} > 0$. $p = cV + b$

Durante la trasformazione B $\rightarrow$ C, p è costante, la trasformazione è isobara e $L = p \Delta V = 30(1-4) = -90 J$

Durante la trasformazione C $\rightarrow$ A, V è costante, la trasformazione è isocora, $L = 0$ e $\Delta E_{int} = Q$

In un ciclo $\Delta E_{int} = 0 \Rightarrow Q = L$

A $\rightarrow$ B: Quando $p = 10 \text{ Pa}$ e $V = 1 \text{ m}^3$ $10 \text{ Pa} = a \cdot 1 \text{ m}^3 + b$ -
 " $p = 30 \text{ Pa}$ e $V = 4 \text{ m}^3$ $30 \text{ Pa} = a \cdot 4 \text{ m}^3 + b =$

$$b = 10 - \frac{20}{3} = \frac{10}{3}$$

$$Q = \frac{20}{3} \text{ Pa} \cdot \text{m}^3 //$$

$$p = \frac{20}{3} V + \frac{10}{3}$$

$$L_{AB} = \int_1^4 p dV = \int_1^4 \left(\frac{20}{3} V + \frac{10}{3} \right) dV = \left(\frac{10}{3} V^2 + \frac{10}{3} V \right) \Big|_{V=1}^{V=4} = \frac{200}{3} - \frac{20}{3} = \frac{180}{3} = 60 \text{ J}$$

$$L_{BC} = -90 \text{ J}$$

$$L_{CA} = 0$$

$$L_{net} = 60 \text{ J} - 90 \text{ J} = -30 \text{ J} = Q_{tot}$$

(46) $L = 200 \text{ J}$ Compiere sul sistema $L < 0$

$Q = 70 \text{ cal}$ Estraggono = cedono $Q < 0$
 $= 293 \text{ J}$

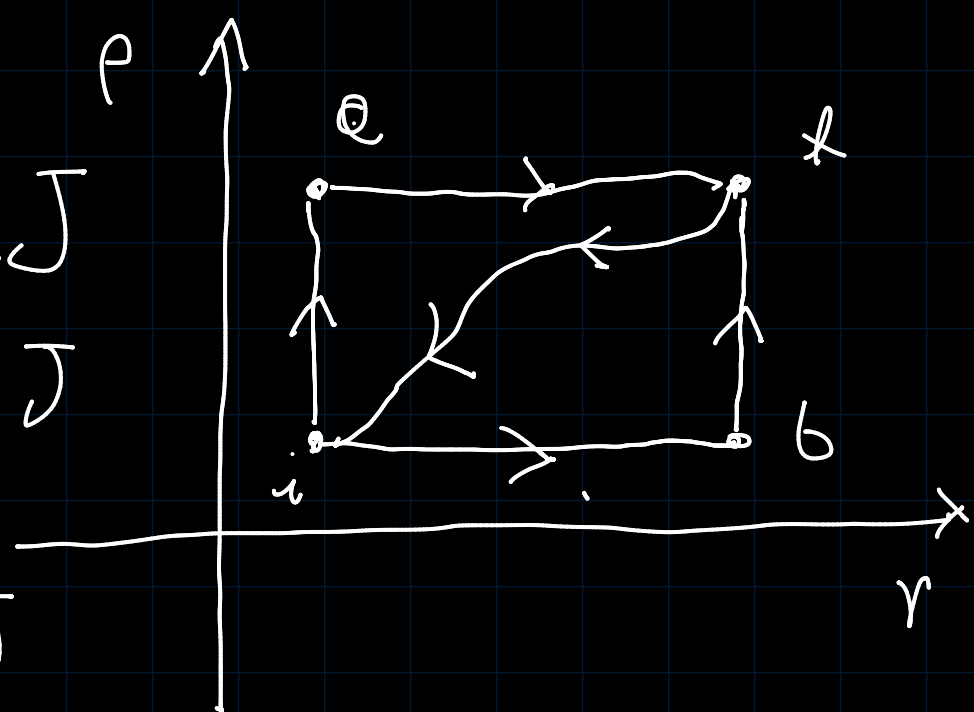
$$\Delta E_{int} = Q - L$$

$$= -Q + L = -293 + 200 = -93 \text{ J} < 0$$

(47) Lungo iaf: $Q = 50 \text{ cal} = 209.2 \text{ J}$

$$L = 20 \text{ cal} = 83.7 \text{ J}$$

Lungo ibf: $Q = 36 \text{ cal} = 150.6 \text{ J}$



$$L_{ibf} = ?$$

Lungo fi : se $L = -13 \text{ cal} = -54.4 \text{ J}$

$$Q = ?$$

Se $E_{int,i} = 10 \text{ cal} = 41.8 \text{ J} \Rightarrow E_{int,f}$

Se $E_{int,b} = 22 \text{ cal} = 92.05 \text{ J}$, Q_{ib} ? Q_{bf} ?

$$\Delta E_{int} = Q - L = 202.2 \text{ J} - 83.7 \text{ J} = 118.5 \text{ J}$$

$$L_{ibf} = Q_{ibf} - \Delta E_{int} = 150.6 \text{ J} - 118.5 \text{ J} = 32.1 \text{ J}$$

Lungo fi il sistema si comprime quindi $\Delta E_{int} < 0$

$$-\Delta E_{int} = Q - L \Rightarrow \Delta E_{int} = L - Q \Rightarrow Q = L - \Delta E_{int}$$

$$= -54.4 \text{ J} - 118.5 \text{ J} =$$

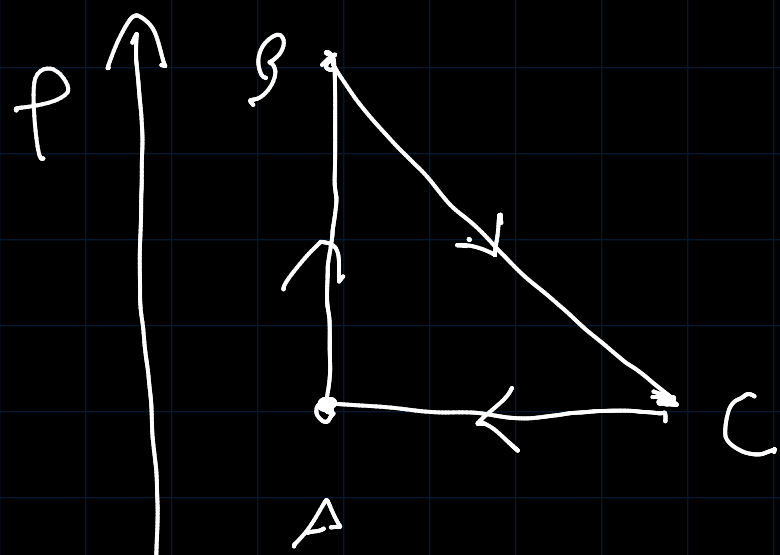
$$= -172.9 \text{ J}$$

$$E_{int,f} = \Delta E_{int} + E_{int,i} = +118.5 \text{ J} + 41.8 \text{ J} = 160.3 \text{ J}$$

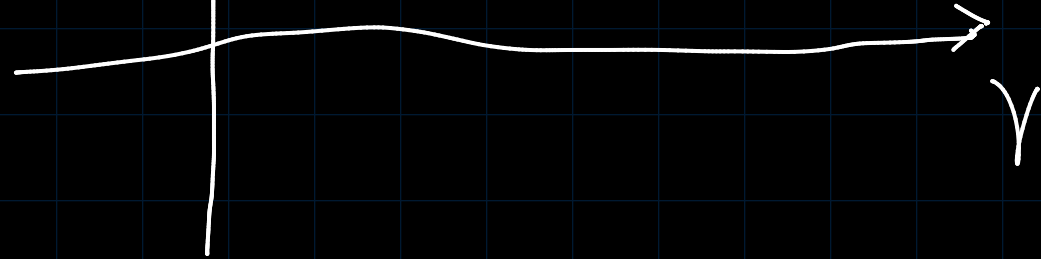
$$Q_{bf} = \Delta E_{int,bf} + L_{bf} = (\Delta E_{int,f} - \Delta E_{int,b}) + 0 \text{ J} = 160.3 \text{ J} - 92.05 \text{ J} = 68.25 \text{ J}$$

$$Q_{ib} = Q_{bf} - Q_{bf} = 150.6 \text{ J} - 68.25 \text{ J} = 82.35 \text{ J}$$

48) $C \rightarrow A$: isobore $= p \text{ const}$
 $A \rightarrow B$: isocore $= V \text{ const} \Rightarrow L = 0$



$$B \rightarrow C: \text{Adiabatica} \Rightarrow Q = 0$$



$$Q_{AB} = 20 \text{ J}$$

$$L_{\text{tot}} = 15 \text{ J}$$

$$Q_{CA}?$$

Durante un ciclo $\Delta E_{\text{int}} = 0 \Rightarrow L = Q$

$$L_{\text{tot}} = Q_{\text{tot}} = Q_{AB} + Q_{BC} + Q_{CA} \Rightarrow$$

$$\Rightarrow 15 \text{ J} = 20 \text{ J} + 0 \text{ J} + Q_{CA} \Rightarrow Q_{CA} = -5 \text{ J}$$

$$50) L_{\text{tot}} = Q_{\text{tot}} = 1,2 \text{ J}$$

$$\Delta E_{\text{int}, ab} = 3 \text{ J}$$

$$|L_{ab}| = 5 \text{ J}$$

$$Q_{ca} = \Delta E_{\text{int}, ca} = 2,5 \text{ J}$$

Im ca $L=0$ perché è una trasformazione isocora

$$Q_{ab} ? Q_{bc} ?$$

$$Q_{ab} = \Delta E_{\text{int}, ab} + L_{ab} = 3 \text{ J} + 5 \text{ J} = 8 \text{ J}$$

$$Q_{\text{tot}} = Q_{ab} + Q_{bc} + Q_{ca}$$

$$1,2 \text{ J} = 8 \text{ J} + Q_{bc} + 2,5 \text{ J}$$

$$Q_{bc} = -10,5 \text{ J} + 1,2 \text{ J} = -9,3 \text{ J}$$

